Practice Questions

INF10025 Data Management and Analytics · Test 3 revision (Weeks 1–8) · every SQL answer was run on a copy of the unit data

Level	Content	Questions
P – Pass	Weeks 1–6 core: concepts, Access criteria, ERD basics, one-table SQL, row counts, error codes	Q1–Q41
C – Credit	Joins, aggregates, GROUP BY / HAVING, constraints, identifiers, weak entities	Q42–Q60
D – Distinction	Week 8: subqueries, outer joins, set operators, ERD → schema, M:M, identifier chains	Q61–Q80
HD – High Distinction	Integrative scenarios, error prediction, design + SQL, code-segment ordering	Q81–Q90
Mock Test 3	12 mixed questions in Test-2 style (suggested time 30 min)	Section 5

How to use

Write each answer yourself first (on paper – Test 3 has no computer help for design questions), then compare with the model answer. Test 3 is worth 5 marks (pass 2.5) and is for Distinction / High Distinction aspirants, so aim to finish levels P and C quickly and spend your time on D and HD.

ERD notation reminder

- **Strong entity** = single box; **weak entity** = double outline. Identifier attributes are underlined.
- **Crow's foot** = MANY end; plain line = ONE end. "|" = **must**, "O" = **may**.
- An underlined relationship name means the entity at the MANY end borrows the identifier of the ONE end (weak entity).
- Foreign keys always go on the MANY side. A "must" at the ONE end → the FK is NOT NULL.
- A M:M relationship is replaced by a weak (intersection) entity with two M:1 relationships; no surrogate keys unless required.

Schema format used in the answers: **TABLE (col1, col2, …) PK (…) FK (…) → PARENT**.

Section 0 – Datasets used in the questions

Row-count questions use the small tables below (copy them into Oracle / Access to check your answers). The movie database (MOVIE 395 rows, ACTOR 826, CASTING 3317, RATING 5, COLOURTYPE 2) is the one from the laboratory.

0.1 Test-2 style tables

CUSTOMER

custno	custname	typecode	sales	location
21	Wendy	1	300.0	Haw
24	Sue	3	700.0	Haw
27	Nick	2	NULL	Kew
22	Dave	4	95.0	Richmond
25	Jenny	1	525.0	Haw
29	Bob	4	15.0	Kew
23	Helen	3	200.0	Haw
28	Linda	1	25.0	Kew
26	Xena	3	NULL	Richmond

TYPE

TypeCode	Title	gradeID	category
1	Under 20	WC	X
2	Under 20	M	Z
3	20 plus	WC	Z
4	20 plus	M	X
5	Special	M	S

GRADE

gradeID	description
WC	Women and Children
M	Men

0.2 Employees and departments

DEPT

DeptNo	DeptName
10	Sales
20	IT
30	HR
40	Legal

EMPLOYEE

EmpNo	EmpName	DeptNo	Salary	Gender	HireDate
1	Anna	10	52000.0	F	2015-03-01
2	Ben	10	48000.0	M	2018-07-15
3	Chloe	20	91000.0	F	2012-01-10
4	Dan	20	78000.0	M	2019-11-30
5	Eve	20	NULL	F	2021-05-02
6	Finn	30	45000.0	M	2020-02-20
7	Gina	NULL	60000.0	F	2022-08-08

0.3 University / lecturer tables

OFFICE

OfficeNo	YearPainted	SqMeters
EN710	2004	12.0
EN505	2011	8.2
BA915	2007	10.5

LECTURER

LecId	LecName	Age	OfficeNo
207	John Smith	37	EN710
119	Jane Pitt	26	EN505
345	Carol Kent	34	EN710
402	Lee Wong	45	NULL

SUBJECT

SubjectCode	Title	CreditPoints	LecId
INF11002	Intro to Web	12.5	207
INF11007	EBIS	12.5	345
INF35700	Honours Project	50.0	207
INF35606	Media Thesis A	25.0	119
INF21009	Advanced Facebook	12.5	NULL

STUDENT

StuId	FirstName	Surname
1122334	Jane	Taylor
4444555	Jobe	McKay
7777700	Helen	Freeman
9900123	Ravi	Patel

ENROLMENT

SubCode	StuId	EnrolDate	FinalGrade
INF10002	1122334	2014-02-28	HD
INF20014	7777700	2014-02-28	P
INF10002	4444555	2014-02-28	C
INF30004	4444555	2014-07-15	P
INF10002	7777700	2014-07-15	HD
INF20014	1122334	2015-02-27	D

UNI_STUDENT

Name
Amanda Allen
Bob Black
Carol Chen
Dave Dole

UNI_TUTOR

Name
Carol Chen
Ed Ellis
Fred Foley

0.4 Campaign and toy-library tables

CAMPAIGN

CampaignCode	CampaignTitle
1	Campaign One
2	Campaign Two
3	Campaign Three
4	Campaign Four

PLAYER

Pid	PName	Gender	CampaignCode
21	Ali Aien	M	2
22	Robert Tipping	M	1
23	Lynda Rogers	F	1
24	Sheelan Aien	F	2
25	Hieu	M	3

PATRON

PatronNo	Firstname	Surname	Gender
53	Sofia	Vergara	Female
71	Alex	Dyson	Male
80	Kim	Lee	Female

TOY

ToyNo	Description	YearAcquired
27	Rocking Horse	2013
28	Jigsaw Puzzle	2015
31	Spinning Top	2014
35	Building Blocks	2016

LOAN

PatronNo	ToyNo	DateBorrowed	DateReturned
53	27	2024-06-01	2024-06-08
53	31	2024-06-01	2024-06-08
53	27	2024-07-19	2024-07-30
53	27	2024-08-02	NULL
71	27	2024-06-11	2024-06-18
71	31	2024-06-11	2024-06-18
71	28	2024-07-31	2024-08-30
53	31	2024-10-09	NULL

Section 1 – Pass (P)

Q1. [Pass] Week 1 – data

Classify each item as structured, semi-structured or unstructured data: (a) a Student table with StudentID, Name, DateOfBirth; (b) an XML document; (c) a folder of customer e-mails and voice messages; (d) a NoSQL document store.

Q2. [Pass] Week 1 – RDBMS

Which statement is NOT an advantage of an RDBMS compared with a spreadsheet?
a) Only the rows that match a search are retrieved
b) The data exists in one place so all users see a change immediately
c) Each user keeps a private copy of the data to work faster
d) Constraints and business rules are easier to enforce

Q3. [Pass] Week 1 – relational model

State the four properties of a two-dimensional table in the relational data model (Codd, 1970).

Q4. [Pass] Week 1 – terminology

Complete the table of equivalent terms: Relation = ____ (Access) = ____ (Oracle); Attribute = ____ (Access) = ____ (Oracle); Tuple = ____ (Access) = ____ (Oracle).

Q5. [Pass] Week 1 – primary key

A table has the column PhoneNo. Explain why PhoneNo is a poor choice of primary key by using the three properties of a primary key from Week 1.

Q6. [Pass] Week 1 – Access views

Give TWO reasons why editing data in Datasheet View is dangerous, and name the Access view where field names and data types are defined.

Q7. [Pass] Week 1 – filters vs queries

State two differences between a datasheet filter and a query.

Q8. [Pass] Week 1 – criteria grid

In the Access design grid the field RELYEAR has criteria 2013 on the “Criteria” row and 2016 on the “or” row. The field RATINGCODE has "M" on the Criteria row and "PG" on the “or” row. Write the equivalent SQL WHERE clause and say how many rows it returns from the MOVIE table.

Q9. [Pass] Week 1 – impossible AND

A student types 2013 And 2016 in the Criteria cell of RELYEAR. How many rows are returned and why?

Q10. [Pass] Week 1 – wildcards

Which titles match the Access criterion Like "the ????" ?
a) The Fall b) The Matrix c) Thelma and Louise d) the Hulk e) The Ring

Q11. [Pass] Week 1 – field names

Why does Access need square brackets in [movie title] but not in movieno? What advice does the lecture give about field names?

Q12. [Pass] Week 2 – FK

The MOVIE table has a column RATINGCODE and a table RATING has RATINGCODE as its primary key. (a) Which column is the foreign key? (b) Which table is the parent? (c) What kind of relationship is it? (d) What does “Enforce Referential Integrity” stop you doing?

Q13. [Pass] Week 2 – Cartesian product

MOVIE has 395 rows and RATING has 5 rows. If the two tables are not joined in a query, how many rows come back? Write the query that shows it.

Q14. [Pass] Week 2 – totals

Write an SQL statement equivalent to an Access Totals query that shows each rating code and the number of movies with that code (heading “Total”).

Q15. [Pass] Week 2 – indexes

(a) What two items does an index store? (b) Name one advantage and one disadvantage of indexing every column. (c) Which index does Access create automatically?

Q16. [Pass] Week 2 – indexes

A table has 100,000 rows and no index on Surname. On average, about how many rows must be read sequentially to find a surname? a) 1 b) 50,000 c) 100,000 d) 2

Q17. [Pass] Week 3 – parameters

Write the criterion you type in the RELYEAR cell so that Access asks the user for a year when the query is run, and name two data types you can choose for a parameter.

Q18. [Pass] Week 3 – action queries

Which of these can be undone with Ctrl+Z after running? Update query, Delete query, Make Table query, Select query. What should you do before running an action query?

Q19. [Pass] Week 3 – import

Explain the difference between importing and linking external data in Access.

Q20. [Pass] Week 3 – Excel dates

How does Excel store the date 2 January 1900, and why may imported Excel dates look like numbers?

Q21. [Pass] Week 4 – Power BI

(a) What does the Relationships window show? (b) Which cross-filter direction should be selected for every relationship? (c) What is the difference between Power BI Desktop and the Power BI web app?

Q22. [Pass] Week 4 – functions

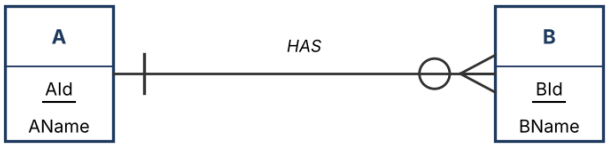
Title = "The Matrix". Give the result of: (a) Left([title],3) (b) Right([title],4) (c) Mid([title],5,3) (d) Ucase([title]) in Access, (e) upper([title]) in Power BI.

Q23. [Pass] Week 5 – ERD concepts

Define: entity, attribute, identifier, business rule. Why is an ERD “a plan” and not a database?

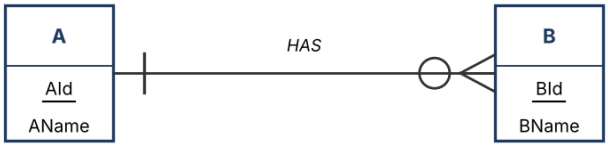
Q24. [Pass] Week 5 – reading ERD

For the ERD below read the relationship in both directions (each sentence must start with the word ONE), and state which table receives the foreign key.



Q25. [Pass] Week 5 – schema

Convert the ERD above into a relational schema. Show primary and foreign keys.



Q26. [Pass] Week 5 – identifier rules

List the four rules for an identifier in an ERD.

Q27. [Pass] Week 5 – NULL

LECTURER(LecId, LecName, Age): lecturer 207 is 37, lecturer 771 Karen Fry has Age = NULL. Does Karen appear in SELECT * FROM lecturer WHERE age > 37? In WHERE age <= 37? In WHERE age IS NULL? Explain.

Q28. [Pass] Week 5 – DDL

Write the CREATE TABLE statements (Oracle) for LECTURER(LecId, LecName, Age) and SUBJECT(SubjectCode, Title, CreditPoints, LecId) with the primary keys, the foreign key, and a constraint that every subject MUST have a convenor.

Q29. [Pass] Week 5 – INSERT

Write INSERT statements: lecturer 974 Robert Tipping whose age is unknown, and the subject INF21009 “Advanced Facebook” (12.5 credit points) convened by lecturer 345. Which common mistake gives ORA-00911?

Q30. [Pass] Week 5 – spot the error

Say whether each statement is valid in Oracle and fix any that are not (table EMPLOYEE(EmpId, EmpName, Gender)):
1 SELECT EmpName, Gender
2 SELECT * FROM Employee ORDER BY EmpName
3 SELECT EmpName FROM Employee WHERE Gender Is M
4 SELECT EmpId, EmpName, Gender WHERE Gender = 'Female' ORDER BY EmpId ASC
5 SELECT LecId LecName Age FROM LECTURER

Q31. [Pass] Week 5 – UPDATE/DELETE

Explain what happens if you run UPDATE employee SET salary = salary + 5000; (no WHERE). Describe the safe way to write a DELETE.

Q32. [Pass] Week 5 – ORDER BY

Write a single statement listing LecName and Age of all lecturers in ascending Age sequence and, within the same age, in descending LecName sequence. Then write another using the column number instead of names.

Q33. [Pass] Week 6 – row counts

Use the CUSTOMER table (custno, custname, typecode, sales, location) supplied with the Test-2 style data (see Section 0.1). How many rows does each statement return?

Q34. [Pass] Week 6 – ORDER BY

Which customer name is listed second by: SELECT custname FROM customer WHERE typecode > 1 ORDER BY custname; ?

Q35. [Pass] Week 6 – write SQL

Write a single SQL statement to list all customer names and sales amounts in descending sales sequence.

Q36. [Pass] Week 6 – write SQL

List the names of customers whose sales are between 25 and 100 inclusive OR between 500 and 900 inclusive.

Q37. [Pass] Week 6 – write SQL

List the customer names that contain the letter N (upper or lower case) and the customer names that have a NULL sales amount – as two separate statements.

Q38. [Pass] Week 6 – write SQL

List each location only once (no duplicates) in alphabetical order. Then list customers whose typecode is 1 or 3 using IN.

Q39. [Pass] Week 6 – UPPER trap

Why does SELECT * FROM customer WHERE UPPER(location) = 'Haw' return 0 rows? Correct it in two ways.

Q40. [Pass] Week 6 – BETWEEN/LIKE

Rewrite WHERE age BETWEEN 30 AND 40 using AND, and WHERE age IN (24,26,28) using OR. Which letters does LIKE 'J%' require?

Q41. [Pass] Week 6 – errors

Match the Oracle error to the situation: (1) inserting a subject whose LecId does not exist in LECTURER; (2) deleting a lecturer who still convenes subjects; (3) inserting NULL into a NOT NULL column; (4) smart quotes; (5) SELECT relyear, COUNT(*) FROM movie;

Section 2 – Credit (C)

Q42. [Credit] Week 6 – join

Write one SQL statement that lists each customer name with the Title of its customer type (tables CUSTOMER and TYPE). How many rows are returned?

Q43. [Credit] Week 6 – join (3 tables)

List customer name, type title and grade description for customers whose grade description is "Women and Children", in alphabetical order of customer name.

Q44. [Credit] Week 6 – join syntax

Rewrite the previous question using the JOIN … ON syntax. Which join conditions belong in ON and which in WHERE?

Q45. [Credit] Week 6 – join (subjects)

List the title of each subject together with the name of the lecturer who convenes it. How many rows are returned, and which subject is missing from the result? Why?

Q46. [Credit] Week 6 – join + ORDER

List lecturer name, office number and year painted for every lecturer (tables LECTURER, OFFICE). Which lecturer is missing and why?

Q47. [Credit] Week 6 – self reference

Show every pair of different players (Pid1, Pid2) who play in the same campaign, listing each pair once only.

Q48. [Credit] Week 7 – aggregates

What does each statement return for the EMPLOYEE table? (a) COUNT(*) (b) COUNT(salary) (c) AVG(salary) (d) SUM(salary)/COUNT(*)

Q49. [Credit] Week 7 – GROUP BY

Write the statement that shows each DeptNo and the average salary of that department, rounded to 0 decimals, sorted by DeptNo. How many groups appear? What happens to Gina?

Q50. [Credit] Week 7 – HAVING

List the departments that have more than one employee, with the employee count.

Q51. [Credit] Week 7 – WHERE vs HAVING

Explain why this statement fails and rewrite it so it lists the departments (deptno) whose average salary is above 50000: SELECT deptno FROM employee WHERE AVG(salary) > 50000 GROUP BY deptno;

Q52. [Credit] Week 7 – movies

How many movies are in each rating? Show RATINGCODE and the count, highest count first.

Q53. [Credit] Week 7 – movies

Show the year, number of movies and average runtime of the years that have at least 15 movies.

Q54. [Credit] Week 7 – join + aggregate

List each rating NAME (table RATING) and the number of movies in it.

Q55. [Credit] Week 7 – DDL constraints

Write ALTER/CREATE statements for: a table PRODUCT(ProdNo number PK, Descr varchar(30) NOT NULL, Price number CHECK Price > 0, Colour varchar(10) CHECK Colour IN ('Red','Blue'), Code varchar(5) UNIQUE).

Q56. [Credit] Week 7 – CHECK/UNIQUE

Table T has UNIQUE(Code) and CHECK(Price > 0). For each insert say whether it succeeds: (a) (1,'A1', 5) (b) (2,'A1', 9) (c) (3, NULL, 7) (d) (4, NULL, 8) (e) (5,'B2', NULL) (f) (6,'C3', 0). Name the error codes.

Q57. [Credit] Week 6 – dates

Write a query listing employees hired after 31 December 2018 using TO_DATE, and the Access equivalent of the date literal.

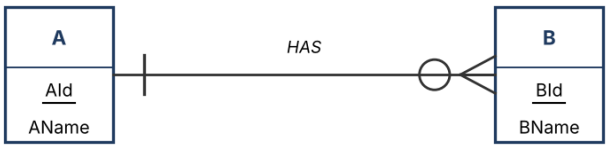
Q58. [Credit] Week 5 – identifier test

The table below holds sample data for entity ENROLMENT. Which attribute(s) form a valid identifier? Test the candidates: (StuId), (SubCode), (StuId, SubCode), (StuId, SubCode, EnrolDate).

SubCode | StuId | EnrolDate | FinalGrade
INF10002 | 1122334 | 28-Feb-2014 | HD
INF20014 | 7777700 | 28-Feb-2014 | P
INF10002 | 4444555 | 28-Feb-2014 | C
INF30004 | 4444555 | 15-Jul-2014 | P
INF10002 | 7777700 | 15-Jul-2014 | HD
INF20014 | 1122334 | 27-Feb-2015 | D

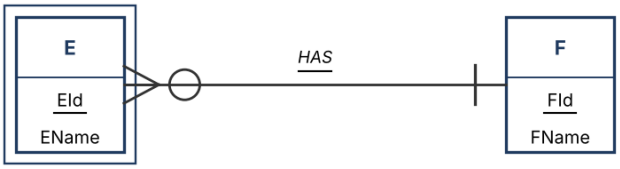
Q59. [Credit] Week 5 – 1:M ERD

Read the ERD below. (a) What is the relationship in words (both directions)? (b) Is B a weak entity? (c) Write the schema.



Q60. [Credit] Week 5 – weak entity

In the ERD below the relationship between E and F is underlined (E has a double outline). State the identifier of E in the relational schema and write the schema.


E
EId
EName
HAS
F
FId
FName

Section 3 – Distinction (D)

Q61. [Distinction] Week 8 – subquery

List the names of employees who earn more than the average salary of all employees.

Q62. [Distinction] Week 8 – subquery IN

List the departments (DeptName) that have at least one employee earning over 60000.

Q63. [Distinction] Week 8 – NOT IN trap

Departments with NO employees. (a) Write it with NOT IN. (b) Explain what happens if the subquery returns a NULL (Gina has NULL DeptNo). (c) Give a safe version.

Q64. [Distinction] Week 8 – outer join

Show every rating name and the number of movies in it, INCLUDING ratings with no movies (count = 0).

Q65. [Distinction] Week 8 – outer join

List every subject title with its convenor name; subjects without a convenor must still appear (show NULL).

Q66. [Distinction] Week 8 – outer join

List all lecturers with the year their office was painted, including lecturers who have no office.

Q67. [Distinction] Week 8 – anti-join

List the players who are NOT in "Campaign Three" without using NOT IN (use an outer join and a NULL test).

Q68. [Distinction] Week 8 – set operators

Table UNI_STUDENT has Amanda Allen, Bob Black, Carol Chen, Dave Dole (4 rows) and UNI_TUTOR has Carol Chen, Ed Ellis, Fred Foley (3 rows). How many rows does each return: UNION, UNION ALL, INTERSECT, MINUS (student MINUS tutor)?

Q69. [Distinction] Week 8 – MINUS in Access

Access has no INTERSECT/MINUS. Rewrite “movies that are rated G but not in colour code 1” style problems (A MINUS B) in a form Access accepts.

Q70. [Distinction] Week 8 – correlated subquery

List employees who earn more than the average salary of THEIR OWN department.

Q71. [Distinction] Week 8 – top rows

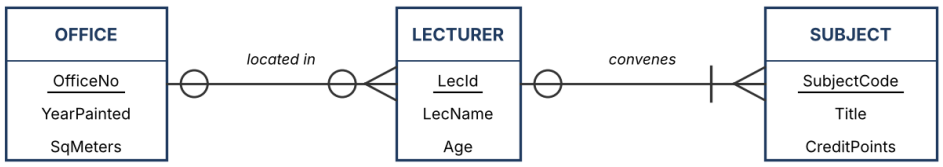
Which rating has the highest average runtime and what is it? Write it with a subquery in HAVING.

Q72. [Distinction] Week 8 – outer join counts

Which actors never appear in CASTING? Write it with a LEFT JOIN and a NULL test, and give the row count.

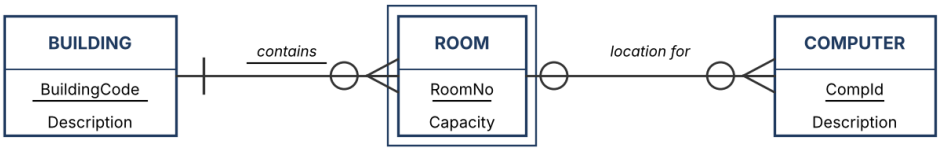
Q73. [Distinction] Week 8 – ERD to schema

Convert the following ERD (OFFICE – LECTURER – SUBJECT) to a relational schema. Where do the foreign keys go? Which foreign keys may be NULL?



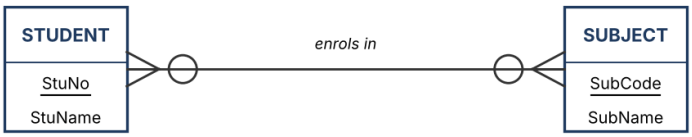
Q74. [Distinction] Week 8 – ERD to schema

Convert the ERD below (BUILDING – ROOM – COMPUTER; ROOM is weak) to a schema. What is the primary key of ROOM and of COMPUTER? Write the foreign key of COMPUTER.



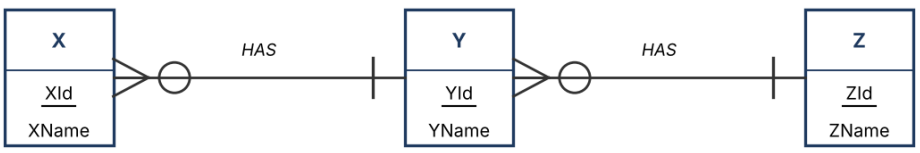
Q75. [Distinction] Week 8 – M:M

The ERD below shows a many-to-many relationship. (a) Why can it not be implemented directly? (b) Redraw it as an ERD with a new entity and give the schema, assuming enrolment date and final grade must be recorded.



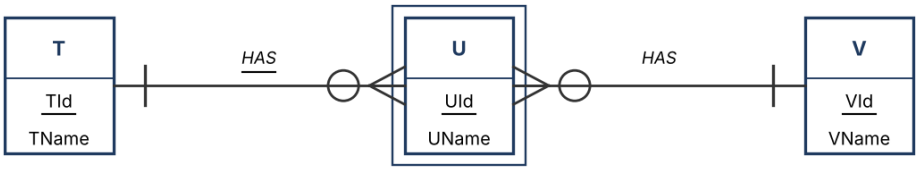
Q76. [Distinction] Week 8 – trace the identifier

Read ERD e3 below (no underlined relationships). What are the primary keys and foreign keys of X, Y and Z?



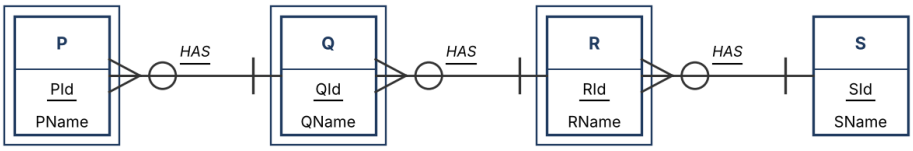
Q77. [Distinction] Week 8 – identifier chain

In the ERD below only the T–U relationship is underlined. Write the PK of U and V and the FKs.



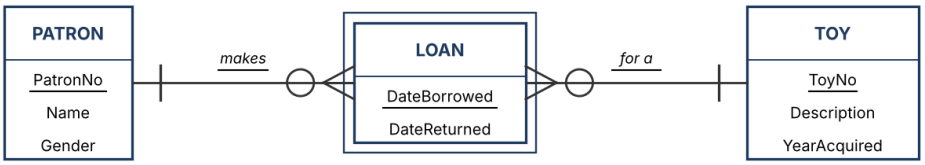
Q78. [Distinction] Week 8 – identifier chain

In the ERD below (e5) all three relationships are underlined. Write the PK of P, Q, R and S.



Q79. [Distinction] Week 8 – design

Toy library: patrons (PatronNo, name, gender) borrow toys (ToyNo, description, year acquired). A loan records date borrowed and date returned (unknown until returned). The same patron can borrow the same toy more than once on different dates. Give the identifier of LOAN, justify it with the sample data, and write the schema.



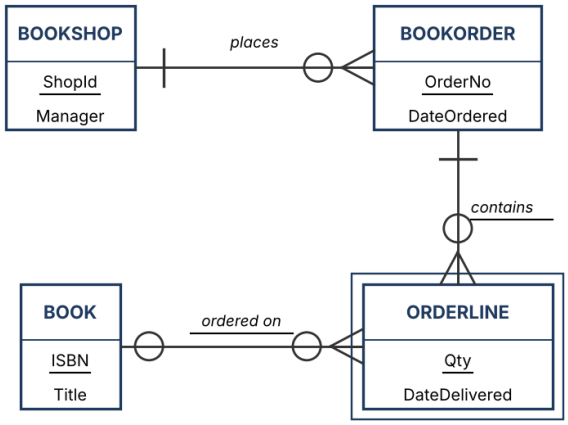
Q80. [Distinction] Week 8 – constraint design

Which of these CHECK constraints are legal in Oracle? (a) CHECK (age >= 0) (b) CHECK (hiredate <= SYSDATE) (c) CHECK (salary > (SELECT AVG(salary) FROM employee)) (d) CHECK (end_date > start_date)

Section 4 – High Distinction (HD)

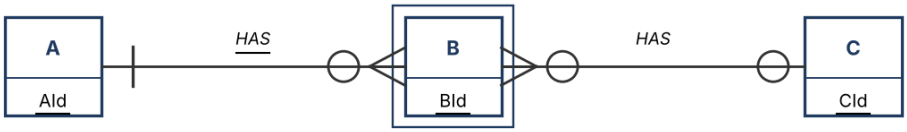
Q81. [High Distinction] Integrative – ERD design

A bookshop chain: each BOOKSHOP (ShopId, Manager) places many orders; an ORDER (OrderNo, DateOrdered) belongs to exactly one shop. Each order contains many order lines, one per BOOK (ISBN, Title) ordered; a line records Qty and DateDelivered (NULL until delivered). A book may be ordered on many orders. Draw the ERD in your head (it is shown below), identify weak entities and write the schema with PKs and FKs.



Q82. [High Distinction] Integrative – ERD critique

The ERD below (A–B and C–B) has underlined relationships on both links. Explain why this is dangerous if C is meant to be a strong entity with its own identifier, and write the schema for both interpretations.



Q83. [High Distinction] Integrative – SQL + join + aggregate

For each campaign title, show the number of players and the number of FEMALE players. Campaigns with no players must still appear.

Q84. [High Distinction] Integrative – SQL

Toy library. List each patron's full name and the number of loans that are still OUTSTANDING (DateReturned IS NULL), including patrons with none. Sort by count descending.

Q85. [High Distinction] Integrative – SQL

Toys that have NEVER been borrowed by patron 71 – give two different SQL answers (NOT IN and outer join).

Q86. [High Distinction] Integrative – SQL

Students (STUDENT) with their highest-level result: list every student's name and the number of subjects they have taken; include students with no enrolments (Ravi). Then list the students who have at least one HD.

Q87. [High Distinction] Integrative – predict

Without running them, decide which statements raise an Oracle error and give the code: (1) SELECT deptno, COUNT(*) FROM employee; (2) SELECT deptno, COUNT(*) FROM employee GROUP BY deptno ORDER BY 2; (3) SELECT empname FROM employee WHERE salary > AVG(salary); (4) DELETE FROM dept WHERE deptno = 10; (assuming FK from employee) (5) INSERT INTO employee (empno, empname) VALUES (8, NULL); (empname NOT NULL).

Q88. [High Distinction] Integrative – mock Test-2 style

Fill in the blanks: A foreign key is placed on the _____ side of a one-to-many relationship; a _____ entity's identifier includes the identifier of its parent; a value of NULL means _____; the clause that filters groups is _____; a join without a join condition produces a _____ product.

Q89. [High Distinction] Integrative – combine code segments

Arrange these fragments into one valid Oracle statement that lists the number of movies per rating for movies longer than 120 minutes where the rating has at least 30 such movies: (i) HAVING COUNT(*) >= 30 (ii) SELECT ratingcode, COUNT(*) (iii) GROUP BY ratingcode (iv) WHERE runtime > 120 (v) FROM movie

Q90. [High Distinction] Integrative – design from scenario

An online course site stores Learners (LearnerNo, Name), Courses (CourseCode, Title) and Certificates. A learner can complete a course several times but never twice on the same day; the completion date and score are stored. Produce the ERD description, the identifier with justification, and the CREATE TABLE statements.

Section 5 – Mock Test 3 (30 minutes)

Attempt without notes. Marks: 12 questions, ½ mark each ≈ 5 marks in total (pass = 5 correct halves). Questions are cross-referenced; answers are in the Answer_Question file under the same numbers.

Mock 1 (Q10) [Pass]

Which titles match the Access criterion Like "the ????" ?
a) The Fall b) The Matrix c) Thelma and Louise d) the Hulk e) The Ring

Mock 2 (Q33) [Pass]

Use the CUSTOMER table (custno, custname, typecode, sales, location) supplied with the Test-2 style data (see Section 0.1). How many rows does each statement return?

Mock 3 (Q41) [Pass]

Match the Oracle error to the situation: (1) inserting a subject whose LecId does not exist in LECTURER; (2) deleting a lecturer who still convenes subjects; (3) inserting NULL into a NOT NULL column; (4) smart quotes; (5) SELECT relyear, COUNT(*) FROM movie;

Mock 4 (Q50) [Credit]

List the departments that have more than one employee, with the employee count.

Mock 5 (Q56) [Credit]

Table T has UNIQUE(Code) and CHECK(Price > 0). For each insert say whether it succeeds: (a) (1,'A1', 5) (b) (2,'A1', 9) (c) (3, NULL, 7) (d) (4, NULL, 8) (e) (5,'B2', NULL) (f) (6,'C3', 0). Name the error codes.

Mock 6 (Q60) [Credit]

In the ERD below the relationship between E and F is underlined (E has a double outline). State the identifier of E in the relational schema and write the schema.



Mock 7 (Q63) [Distinction]

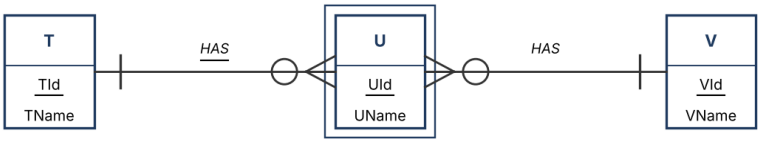
Departments with NO employees. (a) Write it with NOT IN. (b) Explain what happens if the subquery returns a NULL (Gina has NULL DeptNo). (c) Give a safe version.

Mock 8 (Q68) [Distinction]

Table UNI_STUDENT has Amanda Allen, Bob Black, Carol Chen, Dave Dole (4 rows) and UNI_TUTOR has Carol Chen, Ed Ellis, Fred Foley (3 rows). How many rows does each return: UNION, UNION ALL, INTERSECT, MINUS (student MINUS tutor)?

Mock 9 (Q77) [Distinction]

In the ERD below only the T–U relationship is underlined. Write the PK of U and V and the FKs.



Mock 10 (Q84) [High Distinction]

Toy library. List each patron’s full name and the number of loans that are still OUTSTANDING (DateReturned IS NULL), including patrons with none. Sort by count descending.

Mock 11 (Q87) [High Distinction]

Without running them, decide which statements raise an Oracle error and give the code: (1) SELECT deptno, COUNT(*) FROM employee; (2) SELECT deptno, COUNT(*) FROM employee GROUP BY deptno ORDER BY 2; (3) SELECT empname FROM employee WHERE salary > AVG(salary); (4) DELETE FROM dept WHERE deptno = 10; (assuming FK from employee) (5) INSERT INTO employee (empno, empname) VALUES (8, NULL); (empname NOT NULL).

Mock 12 (Q89) [High Distinction]

Arrange these fragments into one valid Oracle statement that lists the number of movies per rating for movies longer than 120 minutes where the rating has at least 30 such movies: (i) HAVING COUNT(*) >= 30 (ii) SELECT ratingcode, COUNT(*) (iii) GROUP BY ratingcode (iv) WHERE runtime > 120 (v) FROM movie